

Technical Publication

Diesel Engine
8V 396 TE 94

Description and
Operation Manual
M010980/10E



Printed in Germany

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Notes for the user

This Manual consists of 10 Sections, marked by tab-index sheets.

These Sections are listed on this page.

Each Section has its own detailed Table of Contents which lists the Groups and Subgroups.

The Groups in Sections B and G correspond to the Groups in the Spare Parts Catalog and other Product Support publications.

Using the Table of Contents to trace an entry

- ➔ Table of Contents (general)
- ➔ Table of Contents (section)
- ➔ Group / Subgroup
- ➔ Heading

Using the Alphabetical Index to trace an entry

- ➔ Term in Alphabetical Index
- ➔ Group / Subgroup
- ➔ Heading

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Please complete and return the Confirmation of Receipt card; you will then receive amendments to your manual.

A Product Summary
Engine summary, technical data

B Description
System and subassembly descriptions

C Operating Instructions
Task descriptions

D Maintenance Schedule
Schedule and maintenance frequency chart

E Troubleshooting
Fault conditions, probable causes, remedies

F Drawings
Engine cross-sections

G Task Description
Inspection and maintenance

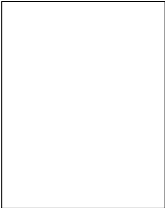
K Preservation
Engine preservation

Z Appendix
Supplementary documentation

? Index
Alphabetical index of key terms

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Diesel Engine

Part A
Product Summary

The product summary gives details of the engine and contains technical data such as engine power, general engine specifications, engine weight, oil and coolant capacities, settings and operational data.

ENGINE LAYOUT, ENGINE POWERKey to Engine Model Designation 8 V 396 TE 94

- 8 = number of cylinders
V = Vee configuration
396 = engine series - 100 times displacement of one cylinder in liters
T = exhaust turbocharging
E = charge air cooling in split-circuit coolant system
9 = engine for marine propulsion
4 = design digit

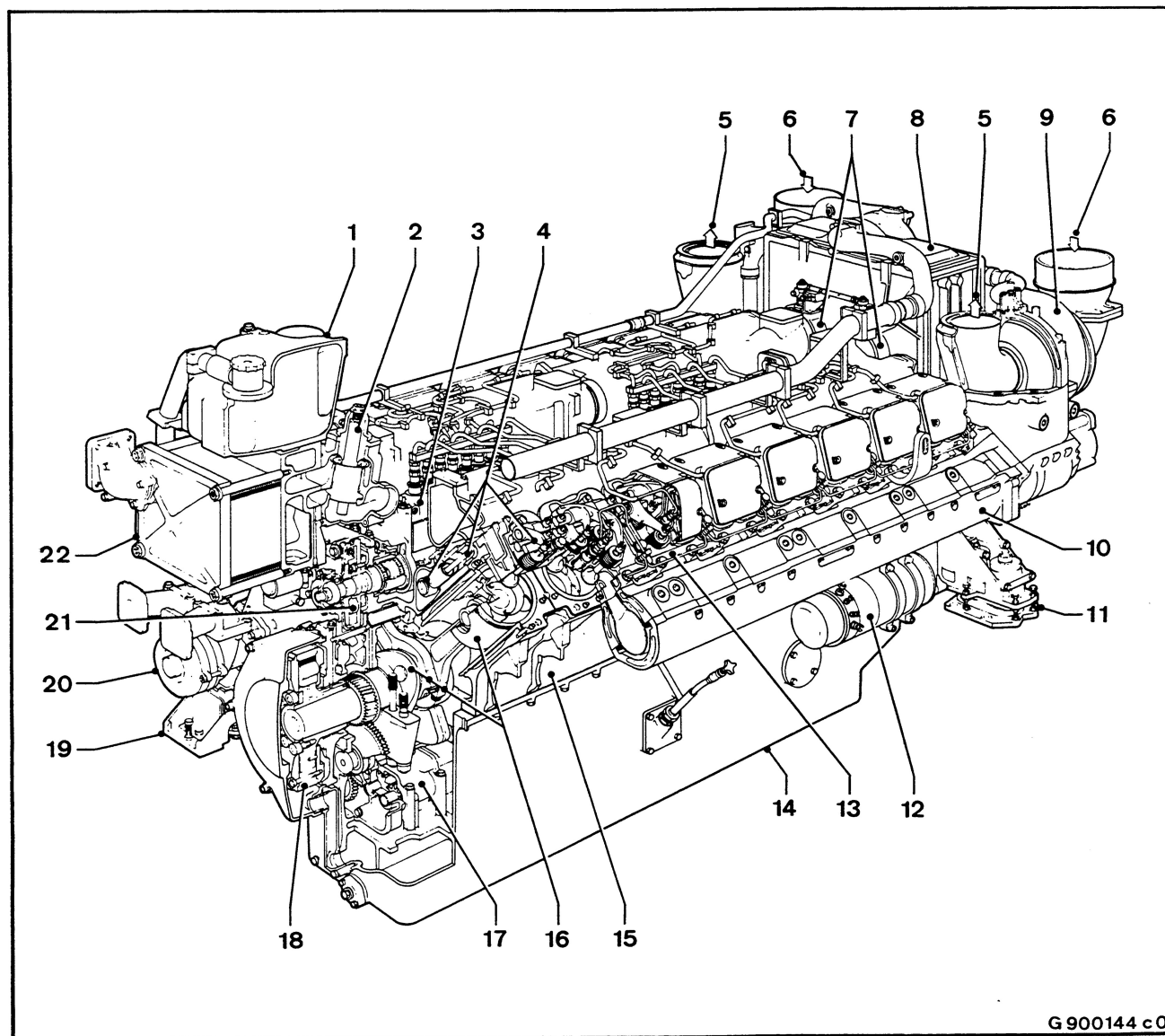
Engine Power

Fuel stop power 1120 kW at 2000 to 2100 rpm

Reference conditions

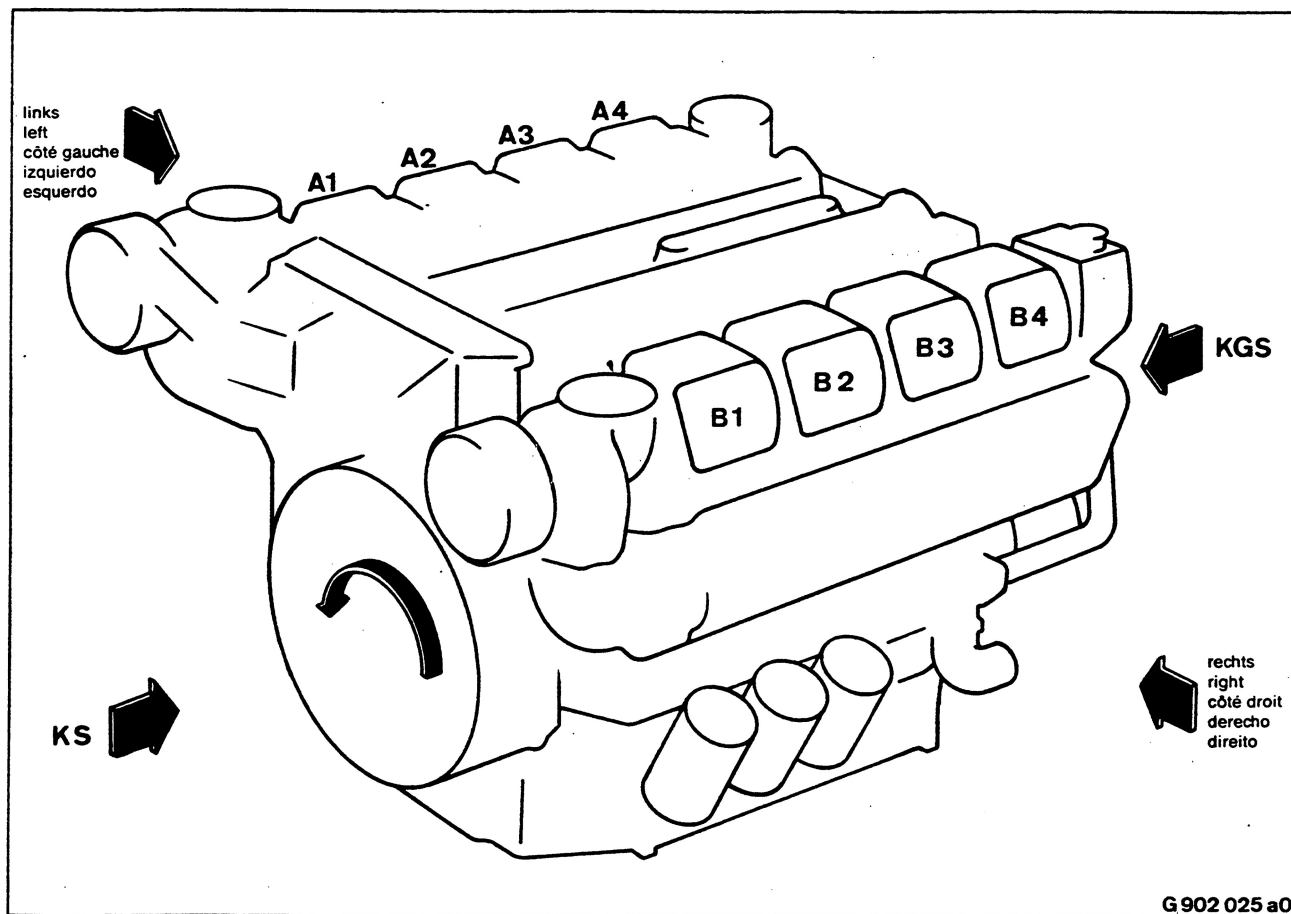
- | | |
|------------------------------------|---------|
| - intake air temperature | 25°C |
| - raw water temperature | 25°C |
| - intake air pressure differential | 20 mbar |
| - exhaust pressure differential | 30 mbar |

Engine Layout Cutaway Drawing 16V 396 TE.4 (also applies to 8 and 12V)



G 900144 c 0

- | | |
|-------------------------------|-----------------------------|
| 1 Coolant expansion tank | 12 Starter |
| 2 Coolant thermostat | 13 Cylinder head |
| 3 Fuel injection pump | 14 Oil pan |
| 4 Valve gear | 15 Crankcase |
| 5 Exhaust outlet | 16 Running gear |
| 6 Air inlet | 17 Engine oil pump |
| 7 Emergency air shut-off flap | 18 Vibration damper |
| 8 Intercooler | 19 Engine mount, timing end |
| 9 Exhaust turbocharger | 20 Bilge pump |
| 10 Exhaust manifold | 21 Fuel injection timer |
| 11 Engine mount, flywheel end | 22 Coolant cooler |

ENGINE SIDE AND CYLINDER DESIGNATIONS

Engine sides are always designated as viewed from the flywheel end.

The left bank of cylinders is marked "A" and the right bank "B" (according to DIN standard 6265).

The cylinders of each bank are numbered consecutively, starting with No. 1 at the flywheel end.

Numbering of other engine components is also from the flywheel end, starting with No. 1.

Designations and abbreviations

Flywheel end = KS (formerly HKS)

Timing end = KGS (formerly GKS)

Left side (formerly A side)

Right side (formerly B side)

GENERAL SPECIFICATIONS

Operating method	four-stroke cycle, single-acting
Combustion method	direct injection
Mode of supercharging	exhaust turbocharging
Cooling method	single-stage sequential turbocharging liquid cooling
Cylinder arrangement	90° Vee
Bore	165 mm
Stroke	185 mm
Displacement, cylinder	3.96 liters
Number of cylinders	8
Displacement, total	31.6 liters
Compression ratio	12.3 : 1
Direction of rotation (viewed from flywheel end)	c.c.w (non-reversible)
Firing order	A1 A4-A2 A3 B4 B3 B2-B1
Fuel injection pressure	
Setting pressure with new spring	280 to 288 bar
Specified pressure with run-in spring	270 to 278 bar
Minimum pressure with run-in spring	260 bar
Final compression pressure at 120 rpm 1000 mbar and engine coolant at 5°C	20 to 24 bar
Break-away torque at engine coolant temperature of 40°C, allowance for acceleration included	approx. 400 Nm
Cranking torque at engine coolant temperature of 40°C	approx. 370 Nm
Firing speed with engine coolant temperature of 40°C	approx. 120 rpm
Mean piston speed	12.95 m/s at 2100 rpm
Engine noise	105 dB (A)
Mean free-field level	1 m distance
Exhaust noise, undamped	121 dB (A)
Free-field level	1 m before outlet